



Configuration Management in the Cloud

What you will learn

At the core of the lesson

You will learn how to:

- Describe the benefits of configuration management
- Identify key AWS services for configuration management
- Create a configuration management strategy

Key terms

- Configuration management



Benefits of configuration management

- Increase efficiency
- Validate every change before release
- Reduce cost by removing unwanted resources
- Enforce security at every layer
- Deploy configuration changes to running instances
- Make configuration automated and repeatable

Management tools



Amazon CloudWatch

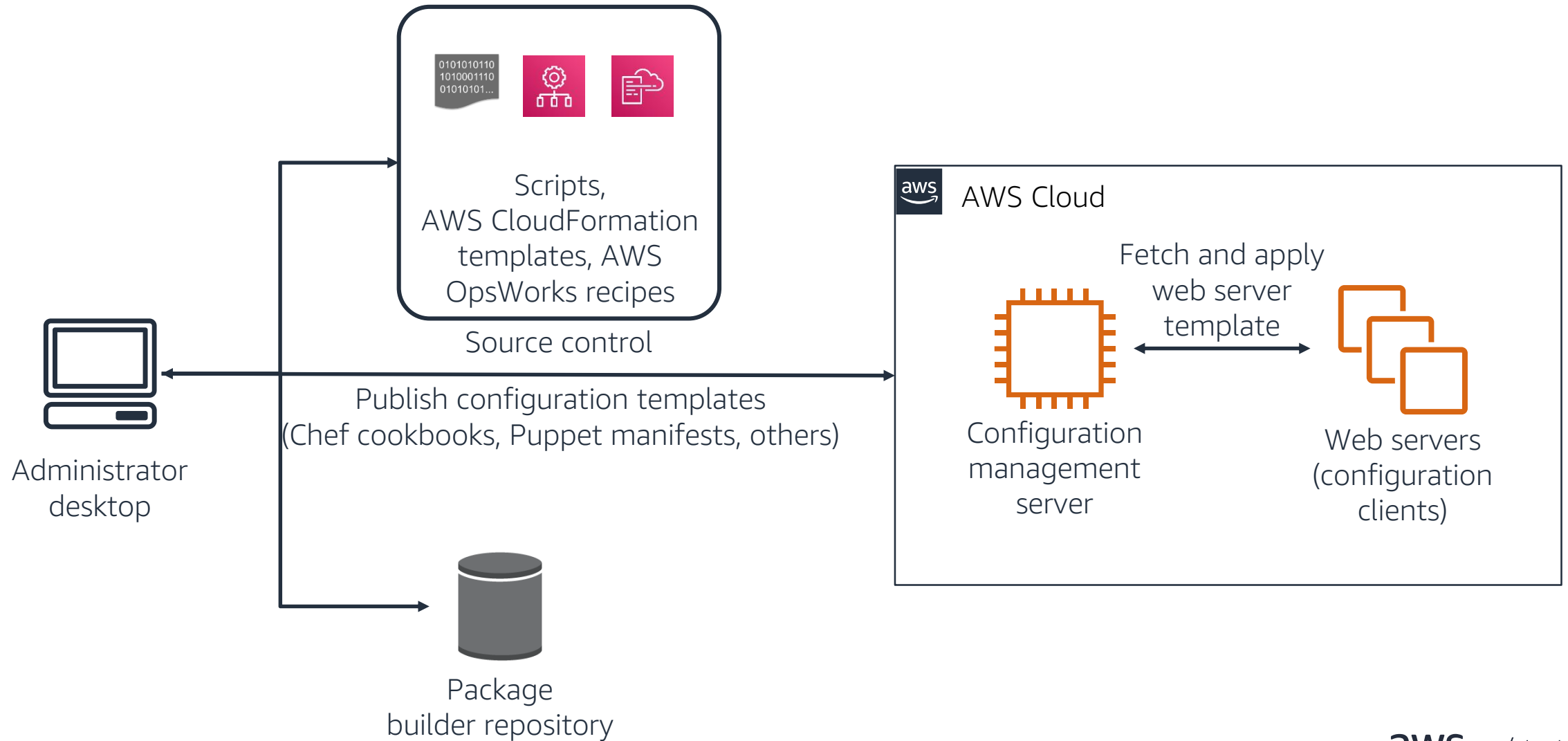


AWS CloudFormation



AWS OpsWorks

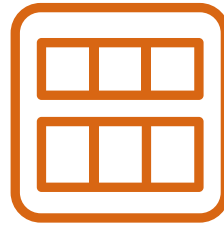
Configuration management tools



Technologies for configuring EC2 instances



User data



Amazon Machine Images (AMIs)



Configuration and deployment frameworks



AWS OpsWorks



AWS CloudFormation

Using configuration software

Benefits of using a configuration server

- Can greatly simplify common administrative tasks.
- Offers a configuration that is idempotent.
 - Resources are allocated only once.
 - Manual changes are detected and rolled back.
- Supply user data to instance to kick off client configuration.
 - Install client and any required configuration and security credentials.
 - Specify the templates to be run.

Creating a configuration strategy

1. Use Amazon Machine Images (AMIs).
 - Foundational components
 - Company-wide tools and standards
2. Use configuration software.
3. Define a source control and change policy.
4. Explore and experiment.

Key takeaways



- Configuration management provides multiple benefits, including validation, automation, documentation, cost reduction, and security.
- AMIs can be pre-created to include required applications and files.
- AWS CloudFormation helps create, update, and delete AWS resources.
- AWS recommends managing a collection of AWS resources as a single unit through AWS CloudFormation.